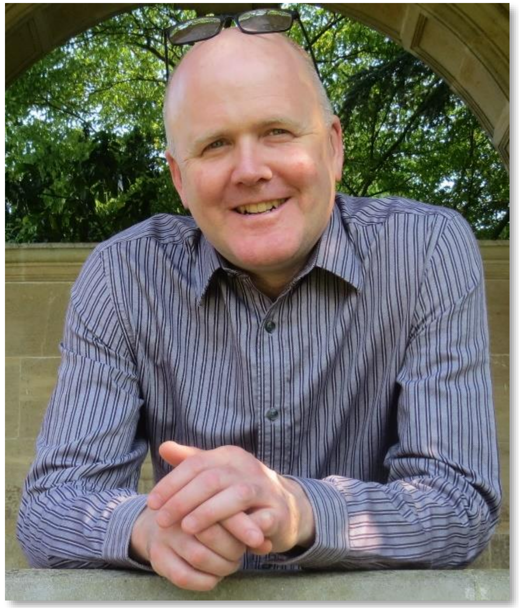
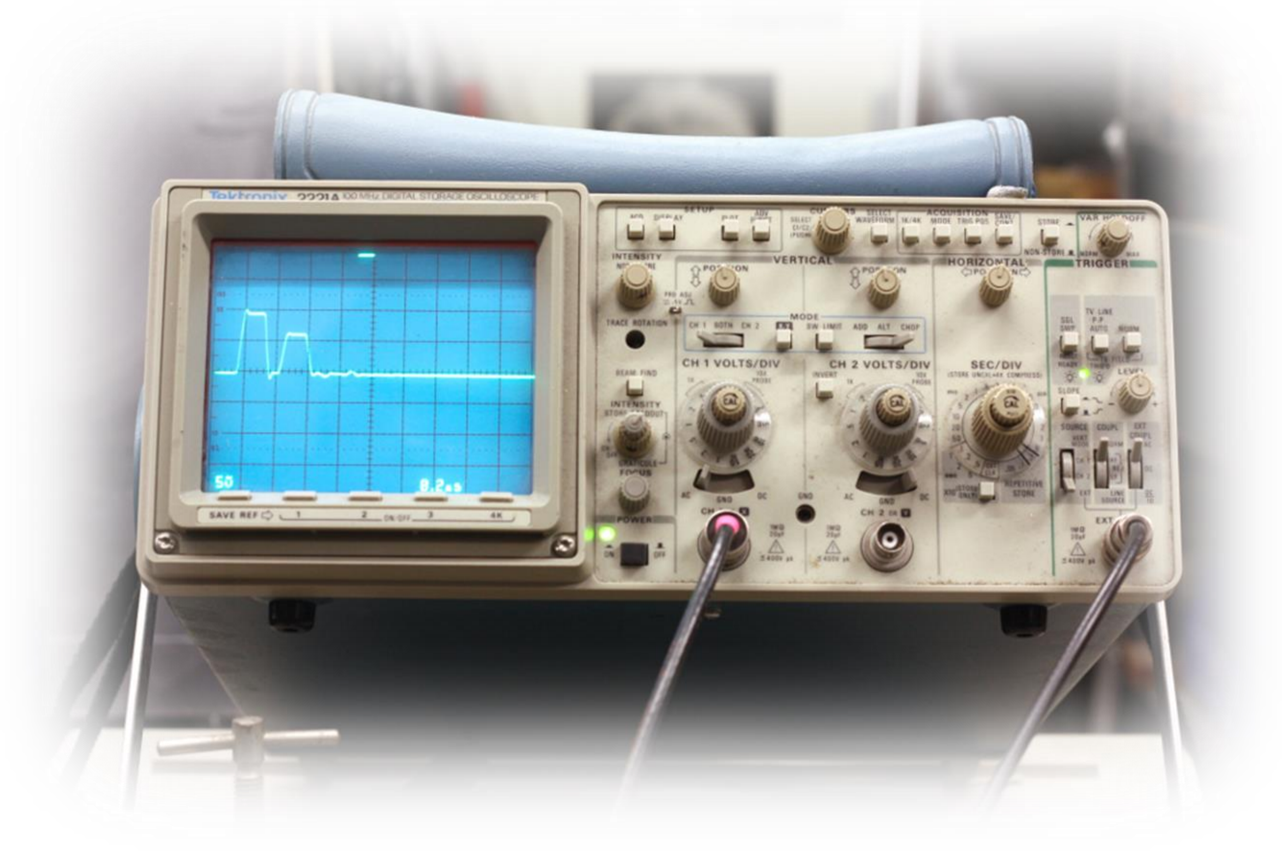
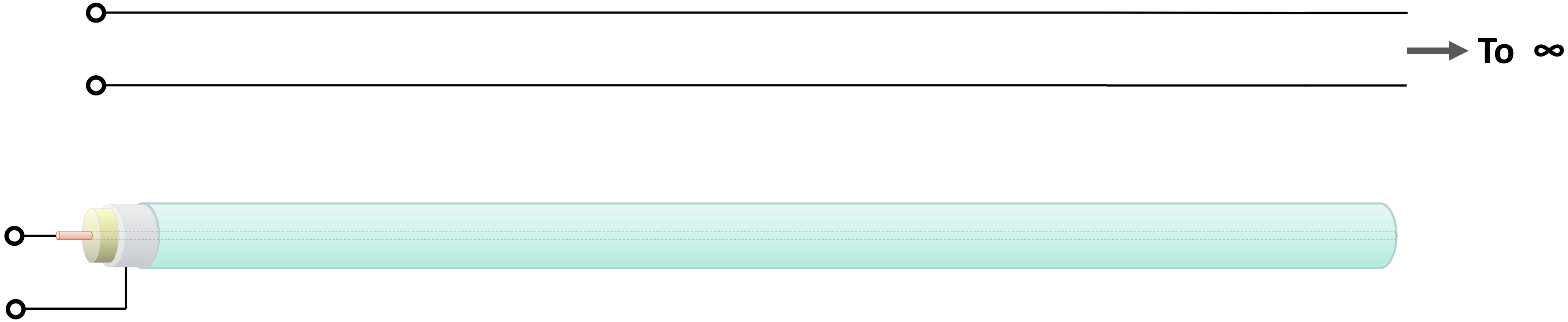
Reflections and Terminations



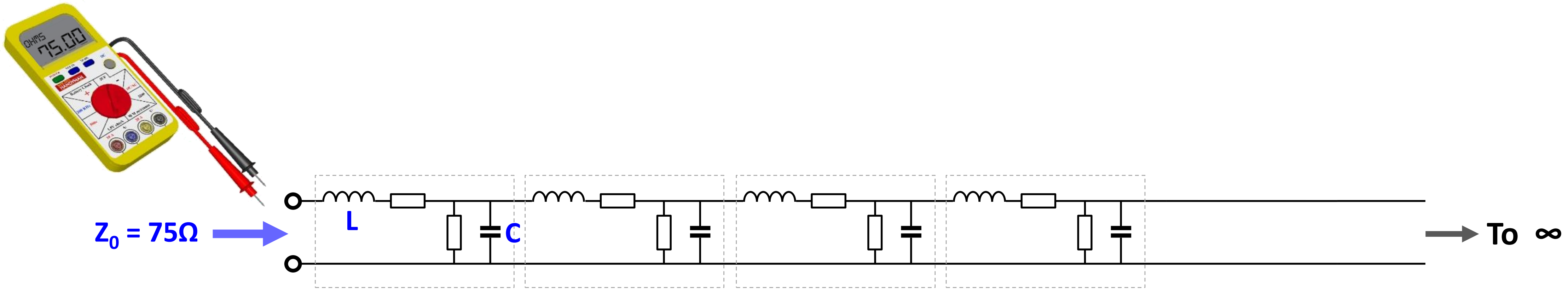


Mike Dhonau

Characteristic Impedance and Terminations



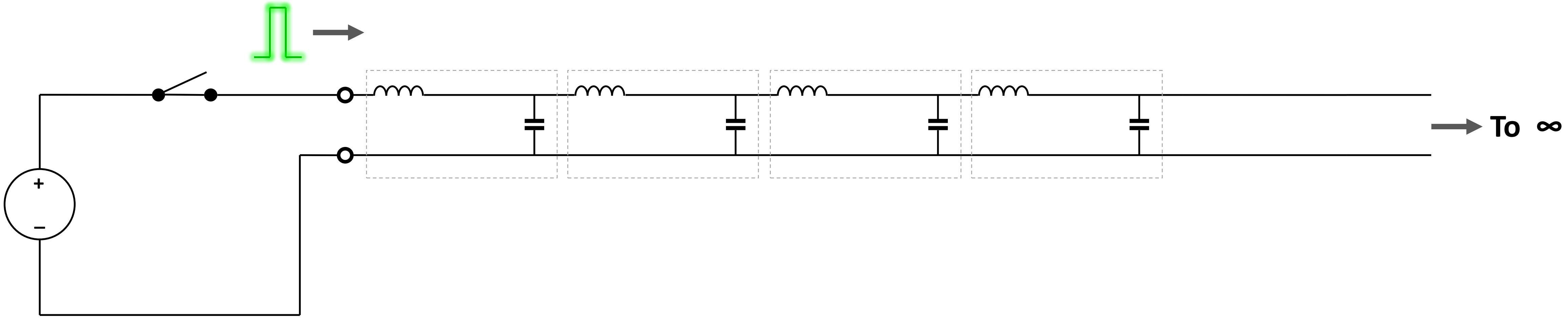
Characteristic Impedance and Terminations



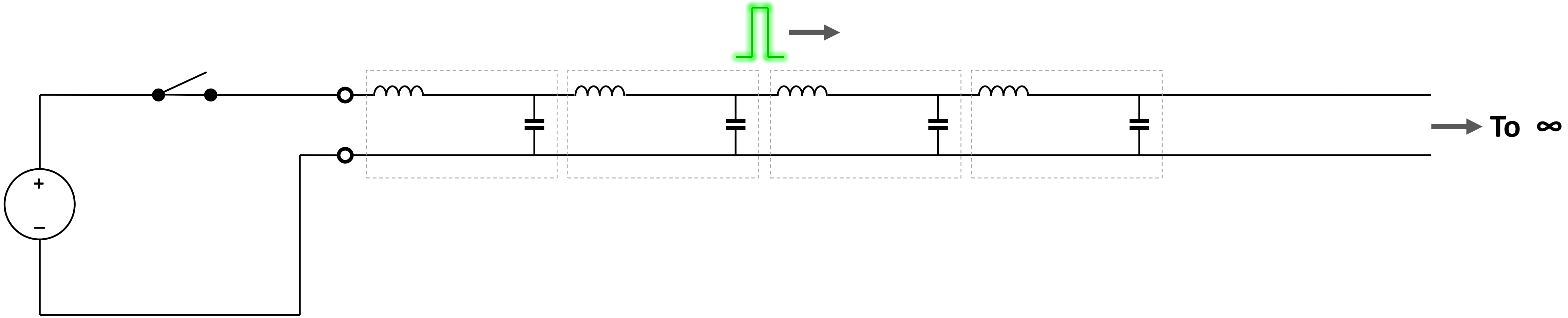
Characteristic Impedance: $$Z_0 = \sqrt{\frac{L}{C}}$$

Characteristic Impedance and Terminations

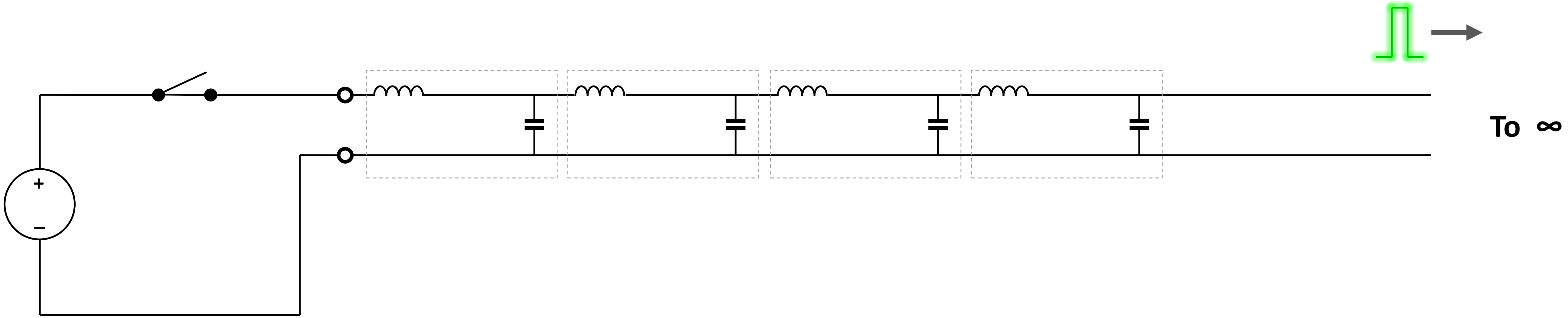
1



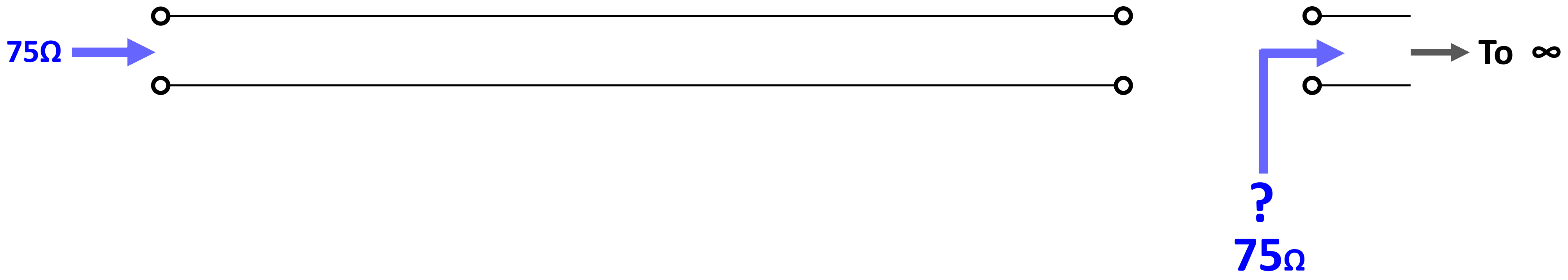
Characteristic Impedance and Terminations



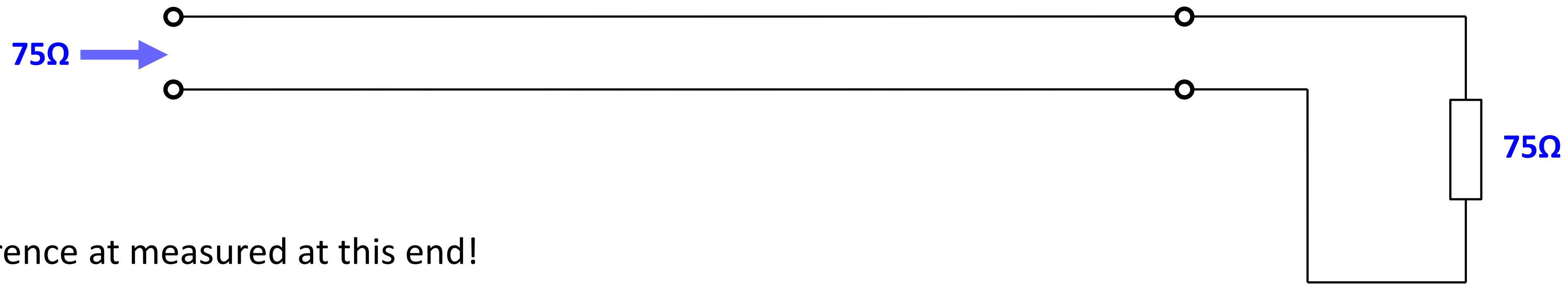
Characteristic Impedance and Terminations



Characteristic Impedance and Terminations



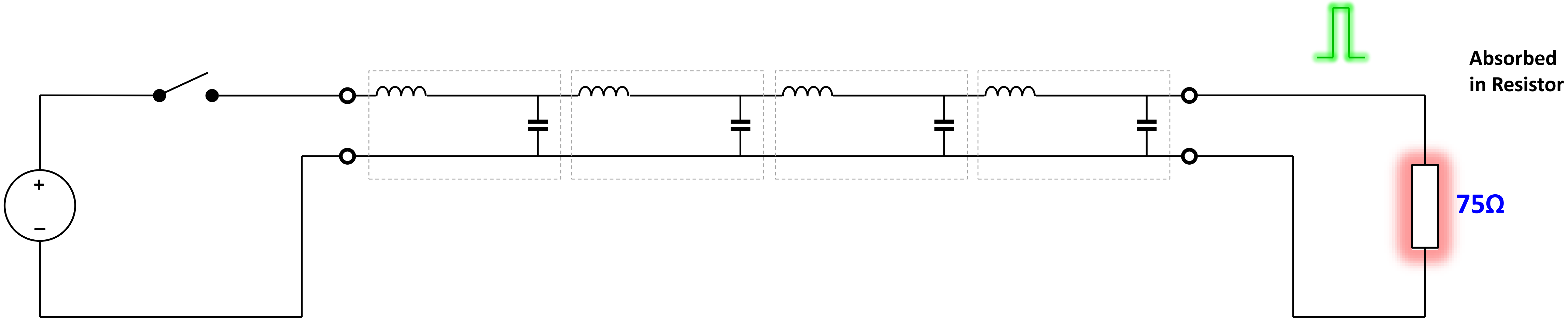
Characteristic Impedance and Terminations



No Difference at measured at this end!

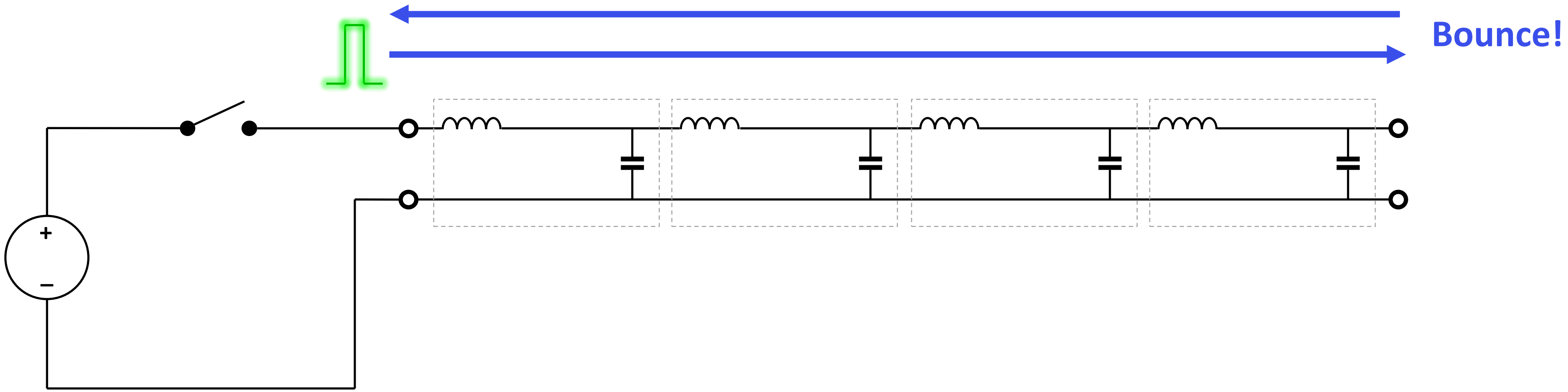
Therefore energy sent still goes to 'infinity'
(is actually absorbed in the termination).

Characteristic Impedance and Terminations



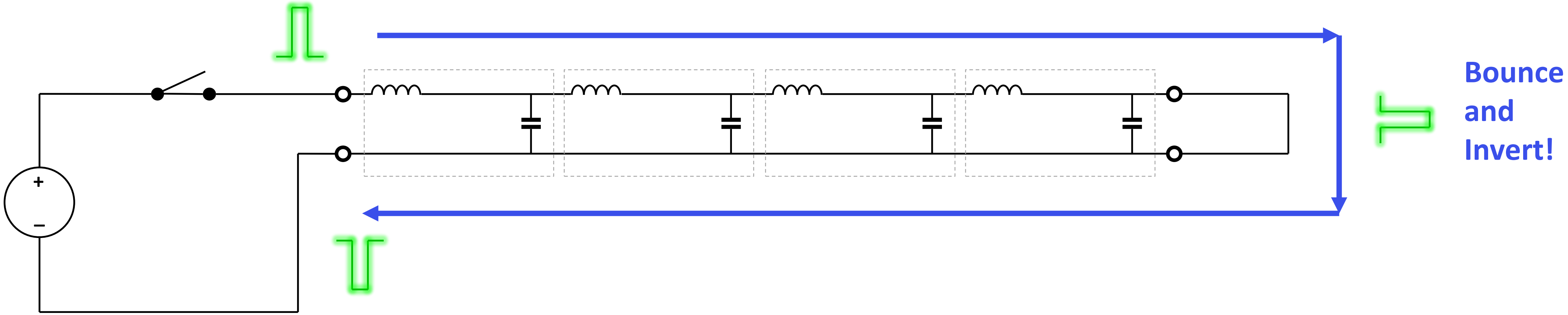
Correct Termination

Characteristic Impedance and Terminations



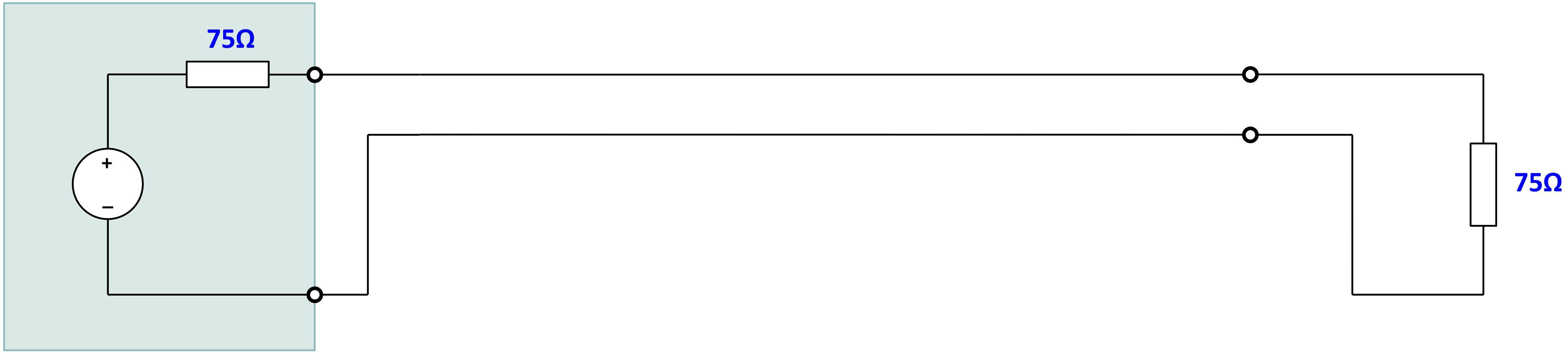
Open Circuit Termination

Characteristic Impedance and Terminations



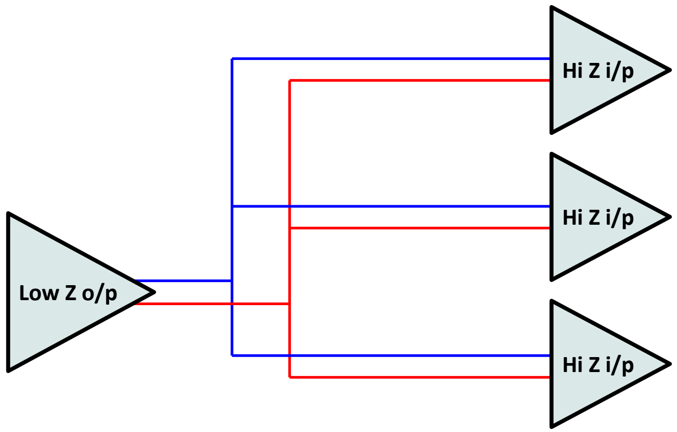
Short Circuit Termination

Correct Termination at Both Ends



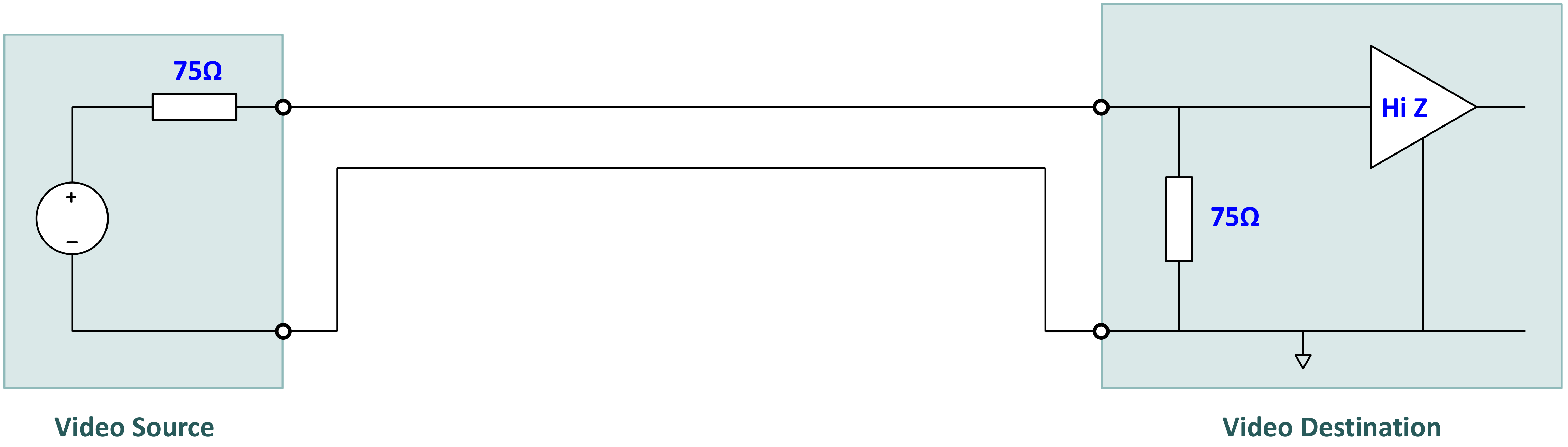
Impedances for Analogue Audio – Audio Distribution

- All professional audio equipment has balanced inputs and outputs.
- Most equipment has the following impedances.
  - High input impedance (typically > 10 kΩ)
  - Low output impedance (typically < 600 Ω)
- Reflections do not matter because the round trip time for audio lines and cables in normal use is very short compared the time for 1 cycle of 20 kHz audio.

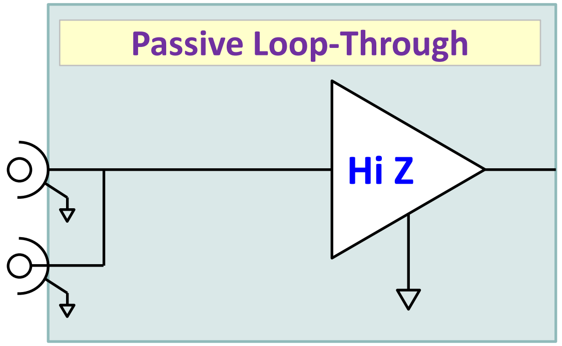


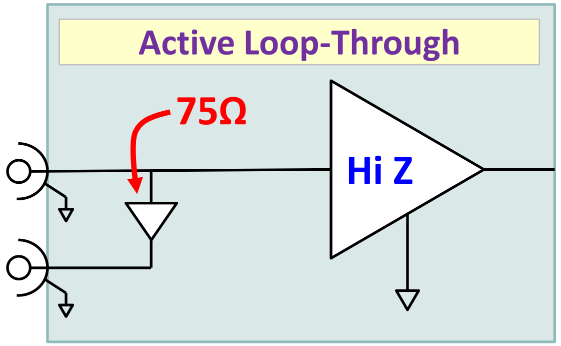
- It is OK to split analogue audio.
  - Sometimes called Parallel / Parallelling
- Long distance (telecoms) audio used 600Ω characteristic impedance cable and correct terminations at each end.
- Such signals can not be paralleled.

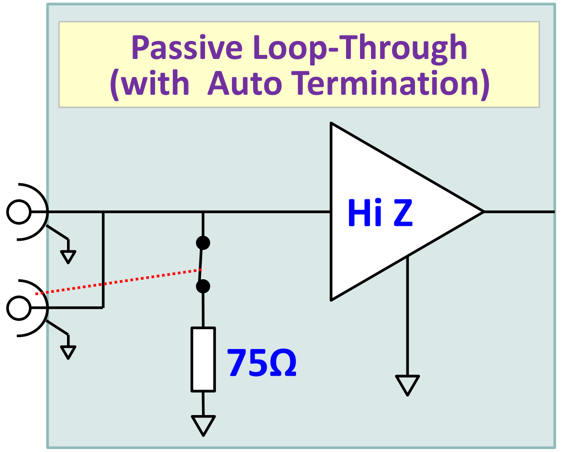
Termination of Video Circuits – Analogue or Video



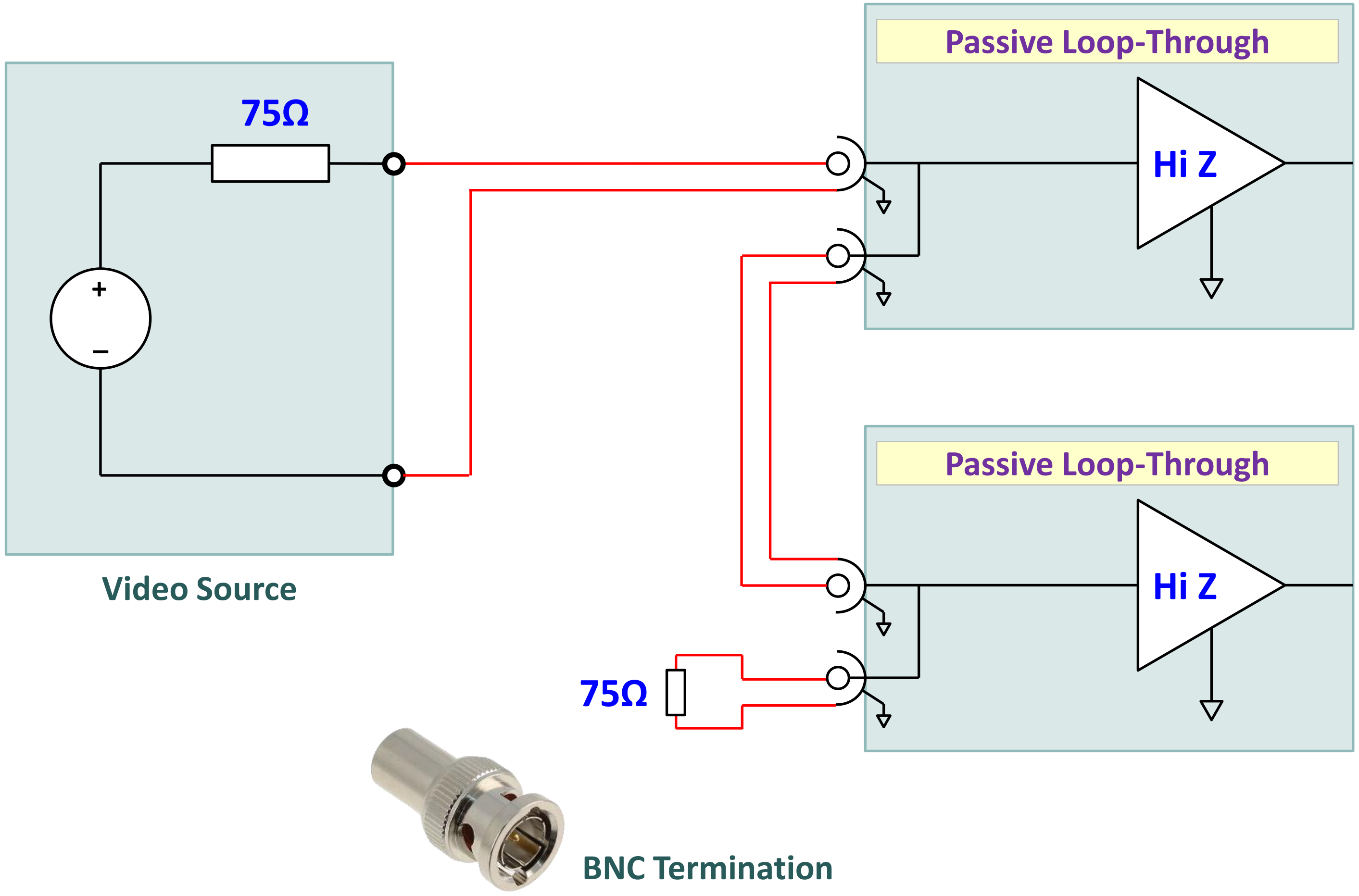
Loop Through Termination



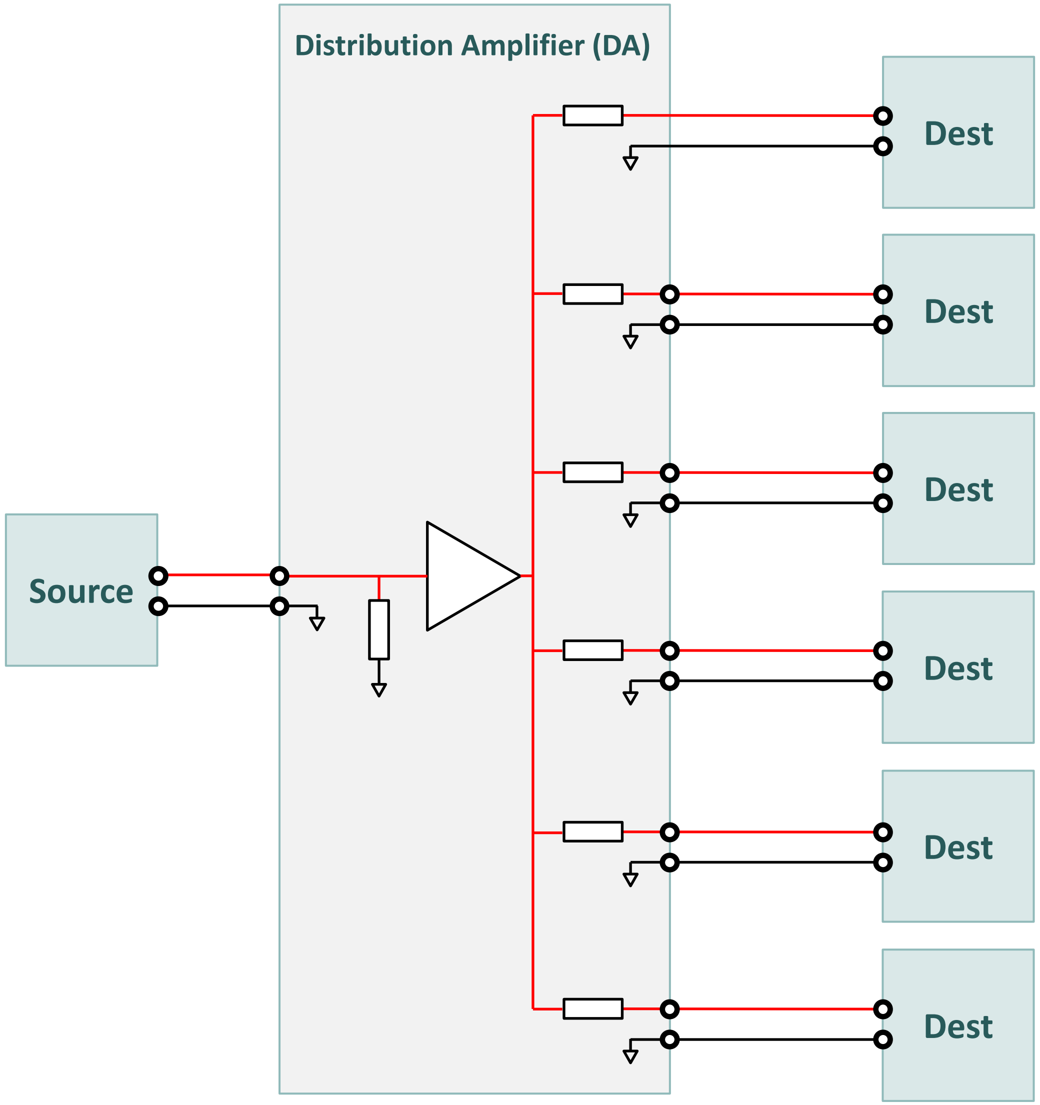




Loop Trough and Termination



Distribution Amps



The End